GOTPM-CPL

(PSC3)

Technical Reference Manual

Hybrid PM + Tree Cosmological
N-Body Simulation Code with
CPL Dark Energy Support

Juhan Kim

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Based on the GOTPM algorithm (Dubinski et al. 2004)

https://github.com/kjhan0606/GOTPM-CPL

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Getting Started

1 Introduction

GOTPM-CPL (Grid-of-Oct-Tree Particle Mesh), also referred to as PSC3 (Parallel Simulation Code 3), is a massively parallel cosmological N-body simulation code that computes the gravitational evolution of dark matter particles in an expanding universe. The code implements the hybrid TreePM algorithm: long-range forces are computed on a uniform mesh using FFT-based Poisson solvers (PM), while short-range corrections are applied through a Barnes–Hut-type tree walk.

1.1 Key Features

- **Hybrid TreePM force calculation** — PM for long-range + Tree for short-range correction
- **CPL dark energy models** — Full support for the Chevallier–Polarski–Linder parameterization $w(a) = w_0 + w_a(1-a)$, including Poisson correction factors
- **2LPT initial conditions** — Built-in 2nd-order Lagrangian perturbation theory for accurate IC generation
- **MPI + OpenMP parallelization** — Slab domain decomposition with hybrid parallelism
- **Variable PM grid (VarPM)** — Adaptive z-domain boundaries for load balancing
- **Index-based position encoding** — High-precision particle positions via displacement + Lagrangian index
- **Custom memory allocator** — Stack-based pool allocator for deterministic memory management
- **Friends-of-Friends (FoF)** — Built-in parallel group finder
- **Power spectrum measurement** — On-the-fly $P(k)$ computation with TSC deconvolution

1.2 Algorithm Overview

The simulation proceeds through the following steps at each time step:

1. **PM force** — Assign particle mass to grid via TSC, solve Poisson equation in Fourier space, interpolate forces back to particles (4th-order FDA).
2. **Tree correction** — Build cell-linked-list, compute short-range force corrections using spline-softened gravity with opening angle θ.
3. **Velocity kick** — Update velocities using leap-frog integration.
4. **Position drift** — Advance particle positions.
5. **Migration** — Exchange particles between MPI domains if they have crossed slab boundaries.
6. **Domain rebalancing** — Adjust slab boundaries if load imbalance exceeds threshold.

Important

The code uses a *split-step* (leap-frog) integrator where velocities and positions are offset by half a time step. The `halfstep` structure controls the synchronization of positions and velocities for output.

2 Quick Start Guide

This chapter provides a step-by-step guide to building and running the code.

2.1 Prerequisites

- **Intel C/Fortran Compilers** — `mpiicx` and `mpiifx` (or `mpiicc/mpiifort`)
- **FFTW3** — Single-precision with MPI and OpenMP thread support (`libfftw3f`, `libfftw3f_mpi`, `libfftw3f_omp`)
- **MPI** — Intel MPI or OpenMPI
- **OpenMP** — Enabled via `-qopenmp`

2.2 Building the Code

Build Commands

```
cd PSC3_CPL_mv_new
# Edit Rules.make for your system (compiler, FFTW path, NMEG)
make clean
make this
# Produces: namu.exe
```

The build system consists of two files:

- `Rules.make` — System-specific compiler settings, library paths, and compile flags.
- `Makefile` — Object file list and build targets.

> The primary target compiler is Intel `mpiicx/mpiifx`. For PGI or GCC compilers, use the corresponding `Rules.make.*` variant (e.g., `Rules.make.pgcc`).

2.3 Setting Up a Run

2.3.1 Directory Structure

Create a separate run directory:

Run Directory Setup

```
mkdir MV_0m0.31_-1.2_-2.4
cd MV_0m0.31_-1.2_-2.4
ln -s ../PSC3_CPL_mv_new/namu.exe .
```

```
cp ../PSC3_CPL_mv_new/params.dat .
# Edit params.dat for your cosmological model
```

2.3.2 Parameter File

Edit `params.dat` to specify cosmological parameters, grid resolution, and run settings. See Chapter 3 for the full parameter reference.

2.3.3 Input Power Spectrum

Prepare an ASCII power spectrum file (e.g., from CAMB) with two columns: k [h/Mpc] and $P(k)$ [$(\mathrm{Mpc}/h)^3$]. Set `Powerflag = 2` and provide the filename in `Ascii Powerfile`.

2.3.4 Job Submission (SLURM)

Example SLURM Job Script

```
#!/bin/bash
#SBATCH --job-name=MV311224
#SBATCH --ntasks=16
#SBATCH --cpus-per-task=4
#SBATCH --time=1000000:00
#SBATCH --output outfile.o%j.grammar
#SBATCH --error errfile.e%j.grammar
#SBATCH -p normal

export OMP_NUM_THREADS=$SLURM_CPUS_PER_TASK
srun ./namu.exe params.dat
```

Choosing MPI Ranks and Threads

The number of MPI ranks (`ntasks`) determines the z-slab decomposition. The FFTW3 domain decomposition requires that N_z is divisible by the number of MPI ranks. Each rank spawns `cpus-per-task` OpenMP threads for shared-memory parallelism within the tree correction and TSC assignment.

II

Code Architecture

3 Parameter Reference

The simulation is controlled by the ASCII parameter file `params.dat`, which is read by rank 0 and broadcast to all MPI processes.

3.1 Cosmological Parameters

Parameter	Default	Description
`OmegaMatter0`	—	Present-day matter density Ω_m
`OmegaBaryon0`	—	Present-day baryon density Ω_b
`OmegaLambda0`	—	Present-day dark energy density Ω_Λ
`Hubble`	—	Hubble parameter $h = H_0/(100\ \mathrm{km/s/Mpc})$
`nPS`	—	Scalar spectral index n_s
`w0 of DE (CPL)`	-1	CPL equation of state w_0
`wa of DE (CPL)`	0	CPL evolution parameter w_a
`Sound Speed of DE`	0	Dark energy sound speed ($c_s^2 = 0$: clustering, 1: smooth)
`fNL`	0	Local non-Gaussianity amplitude
`gNL`	0	Higher-order non-Gaussianity

3.2 Simulation Control

Parameter	Default	Description
`INITIAL CONDITION`	—	1 = Zeldovich approximation, 2 = 2LPT
`Amax`	—	Maximum scale factor $a_{\max}$ (typically 48 for $z_i = 47$)
`Anow`	1	Starting scale factor
`Astep`	—	Scale factor increment per time step Δa
`Nstep`	—	Total number of time steps
`Stepcount`	1	Starting step count (for restart)
`Stepnum`	1	Step number to begin integration
`Theta`	0.3	Tree opening angle (overridden adaptively)
`Iseed`	—	Random seed for IC generation

3.3 Grid and Particle Setup

Parameter	Default	Description
`Nx, Ny, Nz`	—	PM grid dimensions (must be equal: $N_x = N_y = N_z$)
`Nspace`	1	Grid spacing in index units
`Boxsize(Mpc/h)`	—	Comoving box size in Mpc/h

3.4 Power Spectrum and Normalization

Parameter	Default	Description
`Powerflag`	—	0: generate on run, 1: CAMB binary, 2: ASCII file
`Ascii Powerfile`	—	Filename of input power spectrum
`ampRescaling flag`	0	0: none, 1: rescale to σ_8 at $z = 0$, 2: at z_{CMB}
`bias8`	1	Inverse of σ_8 for $R = 8$ Mpc$/h$

3.5 I/O Parameters

Parameter	Default	Description
`rvfilename`	—	Input RV data filename (for restart)
`rvprefix`	—	Output file prefix

4 Compile-Time Flags

The code behavior is controlled by preprocessor flags set in `Rules.make`.

4.1 Required Flags

Flag	Description
`-DINTEL`	Select Intel compiler conventions (main function name). Required for `mpiicx`.
`-DINDEX`	Enable particle index tracking. Each particle stores its Lagrangian grid index for high-precision position reconstruction.
`-DXYZDBL`	Use double-precision position reconstruction via $X = \mathrm{fmod}(x_{\mathrm{disp}} + x_{\mathrm{index}}, N_x)$. Must be paired with `-DINDEX`.
`-DINCLUDE_TREE_FORCE`	Enable tree force corrections on top of PM forces.

4.2 Optional Flags

Flag	Description
`-DNMEG=`*N*`L`	Memory pool size per MPI rank in megabytes (default: 100). Example: `-DNMEG=8000L` allocates 8 GB per rank.
`-DWGROUPSIZE=`*N*	I/O write group size for parallel output (default: 80).
`-DVarPM`	Enable variable-width PM domain decomposition. Allows non-uniform z-slab widths for load balancing.
`-DSAVESLICE`	Save x-z density slices at each time step.
`-DPMSEEDFORCE`	Use PM seed force for initial force evaluation.
`-DMPI_SLOW`	Use slower but more reliable MPI communication patterns.
`-DDEMODEL=0`	Constant Poisson correction factor (default).
`-DDEMODEL=1`	Scale-dependent Poisson correction (for k-dependent DE).
`-DMULTIMASS`	Support for multi-mass particles (adds `mass` field).
`-DINDTIME`	Individual time-stepping support (adds `tflag` field).
`-D_LARGE_FILES`	Enable large file support for I/O.

Example 4.1 (Standard Compilation)

A typical compilation command (via `Rules.make`):

```
CDFLAGS = -DWGROUPSIZE=80 -DNMEG=8000L -DINCLUDE_TREE_FORCE
    \
          -D_LARGE_FILES -DSAVESLICE -DPMSEEDFORCE
COMFLAGS = -DINDEX -DVarPM -DXYZDBL -DMPI_SLOW
OPT = -DINTEL -O2 -qopenmp
```

5 Data Structures

5.1 Particle Types

The code uses two particle structures that share the same memory through pointer casting. This is the core of the hybrid PM+Tree design.

5.1.1 PM Particle (`pmparticletype`)

Definition 5.1 (pmparticletype — 32 bytes)

```
typedef struct pmparticletype {
    float x, y, z;          /* displacement from index
        position */
    float vx, vy, vz;       /* peculiar velocity */
    long long indx;         /* Lagrangian grid index */
} pmparticletype;
```

The PM particle stores only displacement and velocity. The true position is reconstructed via:

$$X = \mathrm{fmod}\Big(x_{\mathrm{disp}} + (\mathrm{indx} \bmod m_x) + N_x,\ N_x\Big)$$

where $m_x = N_x/N_{\mathrm{space}}$. This encoding preserves sub-grid precision even for $N \geq 1024^3$.

5.1.2 Tree Particle (`treeparticletype`)

Definition 5.2 (treeparticletype — 40 bytes)

```
typedef struct treeparticletype {
    float x, y, z;
    float vx, vy, vz;
    long long indx;
    struct treeparticletype *next;  /* linked-list pointer
        */
} treeparticletype;
```

Extends `pmparticletype` with a `next` pointer for the cell-linked-list tree structure. The conversion between the two types is done in-place via backward copy (`pm2treetype`) and forward copy (`tree2pmtype`).

Important

The in-place conversion between `pmparticletype` and `treeparticletype` requires careful handling because both pointers alias the same memory. The code uses `memmove` through `char*` pointers to avoid strict-aliasing violations that can cause data corruption at `-O2` optimization.

5.1.3 Position Reconstruction Macros

```
/* With -DXYZDBL -DINDEX: */
#define XofP(p) fmod((double)((p)->x) + (double)(((p)->indx %
    simpar.mx)) + simpar.nx, simpar.nx)
#define YofP(p) fmod((double)((p)->y) + (double)((((p)->indx %
     simpar.mxmy) / simpar.mx)) + simpar.ny, simpar.ny)
#define ZofP(p) fmod((double)((p)->z) + (double)(((p)->indx /
    simpar.mxmy)) + simpar.nz, simpar.nz)
```

5.2 Simulation Parameters (`SimParameters`)

Definition 5.3 (SimParameters struct)

The global `simpar` structure holds all runtime parameters:

- **MPI identity**: `myid`, `nid`
- **Cosmology**: `omep` (Ω_m), `omeplam` (Ω_Λ), `wlam0` (w_0), `wlam1` (w_a), `CsDE` (sound speed)
- **Grid**: `nx`, `ny`, `nz`, `mx`, `my`, `mz`, `mxmy`
- **Scale factors**: `amax`, `anow`, `astep`
- **Force**: `theta` (opening angle), `pcorr` (Poisson correction)
- **Domain**: `local_nz`, `local_z_start`, `zmin`, `zmax`

5.3 Memory Management

Definition 5.4 (Stack-Based Memory Pool)

The code allocates a single large memory block at startup (`NMEG` megabytes) and manages sub-allocations through a stack-based allocator defined in `Memory2.c`:

- `Malloc(size, PPTR(var))` — Push allocation onto stack
- `Free(ptr)` — Pop or compact stack
- `Realloc(ptr, size)` — Resize with automatic compaction of subsequent allocations
- `freespace()` — Query available memory

The allocator tracks each allocation's size, starting address, and a pointer-to-pointer back-reference, enabling automatic pointer updates when memory is compacted.

The header `Memory.h` defines:

```
#define Realloc(A,B) Realloc(A,B,PPTR(A))
```

> This macro automatically supplies the third argument (pointer-to-pointer), so user code simply calls `Realloc(ptr, newsize)`.

6 Source File Reference

6.1 Core Simulation Loop

File	Description
`main.c`	Entry point. Initializes MPI/FFTW, reads `params.dat`, dispatches to `Zeldovich()` or `SecondOrderLPT()`.
`namu.zeldovich.c`	Main simulation driver for Zeldovich IC. Defines global variables (`DEFINE_SIM_PARA`, `DEFINE_PM_COMMON`, etc.), time integration loop, and type conversion functions (`pm2treetype`, `tree2pmtype`).
`namu.2lpt.c`	Main simulation driver for 2LPT IC. Similar structure to `namu.zeldovich.c` with 2nd-order displacement fields.
`treemain.c`	Tree correction driver. Calls `pm2treetype`, `treecorrection`, `PreFoF`, then `tree2pmtype`.
`pmmain.c`	PM force driver. TSC density assignment, forward FFT, Poisson solve, backward FFT, 4th-order FDA force interpolation.
`dumpmain.c`	Output controller. Triggers snapshot writes when positions and velocities are synchronized.

6.2 Initial Conditions

File	Description
`TwoLPT.F`	2nd-order LPT potential computation in Fourier space. Computes the $\phi^{(2)}$ kernel terms.
`TwoOrderLPT.F`	Converts LPT potentials to particle displacements and velocities.
`addTwoOrderLPT.c`	Applies 2nd-order displacement to particle positions.
`ZeldovichInit.F`	1st-order (Zeldovich) displacement and velocity fields.
`addZeldovichInit.c`	Applies Zeldovich displacement to particles.
`FindSeed.F`	Parallel random number seed initialization.
`sub.lynx.camb.F`	CAMB power spectrum reader and interpolator. Handles fNL non-Gaussianity.

6.3 Force Calculation

File	Description
`tsc.c` / `tsc_omp2.c`	Triangular Shaped Cloud (TSC) mass assignment with OpenMP.
`fft.c`	FFTW3-MPI wrapper for forward/backward transforms. Thread-safe initialization.
`pm_main_sub_fftw3.F`	Low-level FFTW3 workspace management routines.
`p_solver.F`	Poisson solver in Fourier space. Applies $\tilde{\phi}(k) = -\tilde{\rho}(k)/k^2$.
`fda4.c`	4th-order Finite Difference Approximation for force interpolation from grid to particles.
`VarPM.c`	Variable-resolution PM: TSC and FDA with non-uniform z-slab widths.
`Treewalk.c`	Barnes–Hut tree construction and traversal. Builds linked-list octree with opening angle criterion.
`kjhtpm.mod1.gpu.omp.c`	Tree force correction with OpenMP. Cell-linked-list construction (`ParticleCellLinkedList`) and force evaluation (`treecorrection`).
`force_spline.c`	Spline-softened gravitational force kernel.
`pcorr.c`	Poisson correction factor for CPL dark energy models.

6.4 Domain Decomposition and Migration

File	Description
`pmigrate.kisti.omp.c`	MPI particle migration between z-slabs. Handles periodic boundaries and bulk transfer.
`mydomaindecomposition.omp.c`	Adaptive load balancing. Adjusts z-slab boundaries when load imbalance exceeds 1%.
`slicemigrate.c`	FFT data migration between non-uniform and uniform z-slab layouts.
`determine_mpi_misc_param.c`	Computes FFTW3 domain layout parameters (`local_nz`, `local_z_start`).

6.5 Cosmology and Analysis

File	Description
`FLRW.F`	Friedmann equation solver. Growth factors for ΛCDM, quintessence, and CPL models. Functions: `growth()`, `growthcpl()`, `cplOmegaMatter()`, `HofEz()`, `PoissonCor()`.

File	Description
NR.F	Numerical Recipes: Simpson integration (qsimp), Romberg integration (qromo), ODE solver (odeint).
correl.F	On-the-fly power spectrum measurement with TSC window deconvolution.
PreFoF.mod2.omp.c	Parallel Friends-of-Friends group finder preparation (ghost exchange).

6.6 I/O and Utilities

File	Description
kjhrw.c	Parallel binary I/O for particle snapshots (read/write).
header.c	ASCII parameter file reader/writer.
flagwriting.c	Output scheduling: determines when to trigger data writes based on scale factor, step count, and flag files.
Memory2.c	Stack-based memory pool allocator.
timerutil.c	Wall-clock timer utilities.
indexing.c	Particle-to-Lagrangian-grid index mapping.

6.7 Header Files

File	Description
pmheader.h	Particle structures (pmparticletype, treeparticletype), SimParameters, position macros (XofP, YofP, ZofP), timer macros.
pm_common.h	Global PM variables: np, pmparticles, treeparticles, den, work.
flow.h	Evolution factors (EvolFact), half-step structure, output flag function declarations.
Memory.h	Memory allocator API and the Realloc macro.
varpm.h	Variable PM function declarations and NZBound structure.
common_simparam.h	Legacy common-block-style simulation parameters.
fortran.h	C–Fortran interface macros and FFTW/MPI includes.
kjhtree.h	Tree structure definitions.
force_spline.h	Spline kernel function declarations.

III

Algorithms

7 Initial Conditions

7.1 Zel'dovich Approximation (1st-Order LPT)

The Zel'dovich approximation displaces particles from a uniform grid according to the linear displacement field:

$$\mathbf{x}(\mathbf{q},a) = \mathbf{q} - \frac{D_1(a)}{a}\nabla\phi^{(1)}(\mathbf{q}) \tag{7.1}$$

where $\mathbf{q}$ is the Lagrangian coordinate (stored as `indx`), $D_1(a)$ is the linear growth factor, and $\phi^{(1)}$ is the gravitational potential from the linear density field.

The velocity is:

$$\mathbf{v} = \dot{a}\mathbf{x} - a\dot{D}_1\nabla\phi^{(1)} \tag{7.2}$$

7.2 2nd-Order LPT

The 2LPT adds a second-order correction:

$$\mathbf{x}(\mathbf{q},a) = \mathbf{q} - D_1(a)\Psi^{(1)}(\mathbf{q}) + D_2(a)\Psi^{(2)}(\mathbf{q}) \tag{7.3}$$

where $D_2 \approx -\frac{3}{7}D_1^2$ for $\Omega_m = 1$, and

$$\nabla^2\phi^{(2)} = \sum_{i>j}\left(\phi^{(1)}_{,ii}\phi^{(1)}_{,jj} - (\phi^{(1)}_{,ij})^2\right) \tag{7.4}$$

The 2LPT initial conditions are significantly more accurate than Zel'dovich for moderate-to-high redshift starts ($z_i \sim 50$).

Algorithm 7.1 (IC Generation)

1. Generate Gaussian random field $\delta(\mathbf{k})$ using input $P(k)$ and random seed.
2. Compute 1st-order potential: $\tilde{\phi}^{(1)}(\mathbf{k}) = -\tilde{\delta}(\mathbf{k})/k^2$.
3. (2LPT only) Compute 2nd-order source and potential $\phi^{(2)}$.
4. Assign displacements and velocities to particles on a uniform grid.
5. Migrate particles to their correct MPI domains.

8 PM Force Calculation

8.1 TSC Mass Assignment

The Triangular Shaped Cloud scheme assigns particle mass to the $3^3 = 27$ nearest grid points using a quadratic spline window:

$$W_{\mathrm{TSC}}(x) = \begin{cases} \frac{3}{4} - x^2 & |x| \leq \frac{1}{2} \\ \frac{1}{2}\left(\frac{3}{2} - |x|\right)^2 & \frac{1}{2} < |x| \leq \frac{3}{2} \\ 0 & |x| > \frac{3}{2} \end{cases} \tag{8.1}$$

The density field is:

$$\rho(\mathbf{x}_g) = \frac{m_p}{\Delta x^3} \sum_p W_{\mathrm{TSC}}(\mathbf{x}_g - \mathbf{x}_p) \tag{8.2}$$

The implementation in `tsc_omp2.c` uses OpenMP parallelization with thread-local density arrays to avoid race conditions.

8.2 FFT-Based Poisson Solver

Algorithm 8.1 (PM Force Pipeline)

1. **Density field** $\rho(\mathbf{x})$ computed via TSC.
2. **Forward FFT**: $\tilde{\rho}(\mathbf{k}) = \quad [\rho]$ using FFTW3-MPI (`fft.c`).
3. **Poisson solve**: $\tilde{\phi}(\mathbf{k}) = -\tilde{\rho}(\mathbf{k})/k^2$ (`p_solver.F`).
4. **Dark energy correction**: $\tilde{\phi} \to p_{\mathrm{corr}}(a) \cdot \tilde{\phi}$ for CPL models.
5. **Backward FFT**: $\phi(\mathbf{x}) = \quad ^{-1}[\tilde{\phi}]$.
6. **Force interpolation**: 4th-order FDA gives $\mathbf{F} = -\nabla\phi$ at particle positions (`fda4.c`).

8.3 4th-Order FDA

The force at grid point i is:

$$F_i = \frac{1}{12\Delta x}\left(-\phi_{i+2} + 8\phi_{i+1} - 8\phi_{i-1} + \phi_{i-2}\right) \tag{8.3}$$

This requires padding the potential mesh by 3 slices below and 4 slices above the local domain, handled by `paddingpotentialmesh()` in `VarPM.c`.

8.4 Variable PM (VarPM)

With `-DVarPM` enabled, each MPI rank can own a non-uniform number of z-slabs. The z-domain boundaries (`zmin`, `zmax`) adapt to balance particle counts. The TSC and FDA routines accept variable z-boundaries and convert between the local variable-width domain and the global uniform FFTW layout via `tsc_fftw_mesh()` in `VarPM.c`.

9 Tree Force Correction

9.1 Cell-Linked-List Construction

The tree correction uses a cell-linked-list rather than a full octree. In `ParticleCellLinkedList()` (`kjhtpm.mod1.gpu.omp.c`):

1. Divide the local domain into cells of width ~ 4 grid spacings.
2. Assign each particle to its cell using integer truncation of position.
3. Build a linked list through the `next` pointer of `treeparticletype`.

Particle positions near domain boundaries can produce out-of-range cell indices. The code clamps indices to valid range:

```
if(ix < 0) ix = 0;   if(ix >= mxm1) ix = mxm1;
if(iy < 0) iy = 0;   if(iy >= mym1) iy = mym1;
if(iz < 0) iz = 0;   if(iz >= mzm1) iz = mzm1;
```

9.2 Force Evaluation

For each particle, the tree correction computes the short-range gravitational force from nearby particles using a spline-softened kernel. The opening angle criterion θ determines whether to use a cell's center of mass or open it:

$$\frac{\ell}{r} < \theta \quad \Rightarrow \quad \text{use cell monopole} \tag{9.1}$$

where ℓ is the cell size and r is the particle–cell distance. The opening angle is adaptively adjusted during the simulation:

```
if(stepcount < 10)        theta = 0.3;
else if(stepcount < 20)   theta = 0.35;
else if(stepcount < 60)   theta = 0.4;
else                      theta = 0.45;
```

9.3 Type Conversion

Before tree correction, particles are expanded in-place from 32-byte `pmparticletype` to 40-byte `treeparticletype` (`pm2treetype`), and shrunk back afterward (`tree2pmtype`).

Algorithm 9.1 (pm2treetype — In-Place Expansion)

1. `Realloc` the particle array to $N_p \times 40$ bytes.
2. Copy backwards (from last particle to first) using `memmove` to avoid overwriting:

```
for(i = nplong-1; i >= 0; i--) {
    memmove((char*)treeparticles + i*sizeof(
        treeparticletype),
            (char*)pmparticles + i*sizeof(pmparticletype
                ),
            sizeof(pmparticletype));
}
```

3. The `next` pointer (last 8 bytes) is left uninitialized and set during linked-list construction.

10 Time Integration

10.1 Leap-Frog Scheme

The code uses a kick-drift-kick leap-frog integrator where the scale factor a advances by a fixed step Δa:

$$v_{n+1/2} = v_n + \frac{\Delta a}{2}\dot{v}_n \quad \text{(kick)} \tag{10.1}$$

$$x_{n+1} = x_n + \Delta a\, v_{n+1/2} \quad \text{(drift)} \tag{10.2}$$

$$v_{n+1} = v_{n+1/2} + \frac{\Delta a}{2}\dot{v}_{n+1} \quad \text{(kick)} \tag{10.3}$$

The velocity factors `fact1` and `fact2` encode the cosmological dependence:

$$\ddot{a} = -\frac{4\pi}{3}\left(\frac{1}{a^2} + (1+3w)\frac{\Omega_\Lambda}{\Omega_m}\frac{f_{\mathrm{DE}}(a)}{a^3}\right) \tag{10.4}$$

computed by `appapsq()` in `FLRW.F`.

10.2 Half-Step Synchronization

The `halfstep` structure controls when positions and velocities are synchronized (both at the same time level) for output:

- `halfstep.first = 1, second = 0`: First half-step (velocity kick only).
- `halfstep.first = 1, second = 1`: Second half-step (complete the full step).

  This is triggered by output flags (`flagsyncpdata`) and the PreFoF group finder.

11 Cosmology: CPL Dark Energy

11.1 CPL Parameterization

The equation of state for dark energy is parameterized as:

$$w(a) = w_0 + w_a\,(1 - a/a_{\max}) \tag{11.1}$$

where $a_{\max}$ is the maximum scale factor. The standard ΛCDM model corresponds to $w_0 = -1$, $w_a = 0$.

11.2 Expansion History

The Hubble parameter $H(a)$ in a flat universe:

$$H^2(a) = H_0^2 \left[\Omega_m \left(\frac{a_{\max}}{a}\right)^3 + (1 - \Omega_m - \Omega_\Lambda) \left(\frac{a_{\max}}{a}\right)^2 + \Omega_\Lambda\, f_{\rm DE}(a) \right] \tag{11.2}$$

where

$$f_{\rm DE}(a) = \left(\frac{a}{a_{\max}}\right)^{-3(1+w_0+w_a)} \exp\left[-3w_a \left(1 - \frac{a}{a_{\max}}\right)\right] \tag{11.3}$$

is the dark energy scaling function, implemented as `scaleDE()` in `FLRW.F`.

11.3 Growth Factor

The matter growth factor $D_+(a)$ satisfies:

$$\ddot{D}_+ + 2H\dot{D}_+ = \frac{3}{2}\Omega_m(a)\,H^2\,C_w(a)\,D_+ \tag{11.4}$$

where $C_w(a) = 1 + (1+w)\,\Omega_{\rm DE}(a)/\Omega_m(a)$ encodes the dark energy clustering. This is solved numerically via the RK4 integrator in `growthcpl()` with the `odeint` driver from Numerical Recipes.

11.4 Poisson Correction

For dark energy models with clustering ($c_s = 0$), the Poisson equation is modified:

$$\nabla^2\phi = 4\pi G\,\bar{\rho}_m\,\delta_m\,p_{\rm corr}(a) \tag{11.5}$$

where

$$p_{\rm corr}(a) = 1 + \frac{\Omega_{\rm DE}(a)\, D_{\rm DE}(a)}{\Omega_m(a)\, D_m(a)} \tag{11.6}$$

This correction is computed at each time step by `PoissonCor()` in `FLRW.F` and applied in `pmmain.c`.

For smooth dark energy ($c_s = 1$), $p_{\rm corr} = 1$ (no clustering correction).

12 Domain Decomposition and Migration

12.1 Slab Decomposition

The simulation box is divided into N_{rank} slabs along the z-axis. Each MPI rank owns a contiguous range of z-coordinates $[z_{\min}, z_{\max})$. This aligns with the FFTW3-MPI data layout, which distributes the first (z) dimension across ranks.

12.2 Particle Migration

After positions are updated, particles that have moved outside their domain are exchanged via `migrate()` in `pmigrate.kisti.omp.c`:

Algorithm 12.1 (Particle Migration)

1. Scan all local particles; classify as staying, moving up, or moving down.
2. Send "up" particles to rank+1, receive from rank−1 (periodic).
3. Send "down" particles to rank−1, receive from rank+1 (periodic).
4. Compact local array and append received particles.
5. If multi-slab migration is needed, repeat for larger shifts.

12.3 Load Balancing

The domain rebalancer in `mydomaindecomposition.omp.c` monitors the particle count ratio:

$$R = \frac{N_{\max} - N_{\min}}{N_{\text{avg}}} \times 100\% \tag{12.1}$$

If $R > 1\%$ (the `MAXINHOMO` threshold), z-slab boundaries are adjusted so each rank holds approximately equal numbers of particles.

IV

Reference

13 Troubleshooting

13.1 Common Issues

13.1.1 Segfault in Tree Correction

Symptom: Crash in `treecorrection()` or `ParticleCellLinkedList()`.

Possible causes:

1. **Strict aliasing violation in** `pm2treetype`: At `-O2`, the compiler may reorder reads/writes of `pmparticles[i]` and `treeparticles[i]` because they are different types. The fix is to use `memmove` through `char*` pointers.
2. **Out-of-bounds cell index**: Particles near domain boundaries can produce negative or overly large cell indices. Ensure clamping is applied.
3. **Insufficient memory**: If `NMEG` is too small, the `Realloc` in `pm2treetype` fails.

13.1.2 Malloc Error (Extremely Large Allocation)

Symptom: "Error allocating memory N bytes in Malloc()" where N is petabytes.

Cause: OpenMP clamping bug. Check that the y and z index limits use `mym1`/`mzm1` (not `mxm1`) in `ParticleCellLinkedList()`.

13.1.3 Power Spectrum File Not Found

Ensure the `Ascii Powerfile` path in `params.dat` is relative to the run directory, not the source directory.

13.2 Performance Tuning

- **NMEG**: Set large enough to hold all particles plus FFT workspace. Rule of thumb: NMEG $\geq 40 \times N_p^{1/3}$ MB per rank for N_p total particles.
- **MPI vs OpenMP balance**: Use fewer MPI ranks with more threads per rank to reduce communication overhead. Typically 4–8 threads per rank.
- **Opening angle θ**: Larger values are faster but less accurate. The adaptive scheme (0.3–0.45) is a good default.

14 Example: CPL Dark Energy Simulation

This chapter walks through a complete simulation with CPL parameters $\Omega_m = 0.31$, $w_0 = -1.2$, $w_a = -2.4$.

14.1 Parameter File

params.dat

```
#Start of the Ascii Header of the GOTPM Simulation
define INITIAL CONDITION    = 2
define OmegaMatter0         = 0.31
define OmegaBaryon0         = 0.044
define OmegaLambda0         = 0.69
define Hubble               = 0.72
define nPS                  = 0.96
define Boxsize(Mpc/h)       = 1024
define w0 of DE (CPL)       = -1.2
define wa of DE (CPL)       = -2.4
define Sound Speed of DE    = 0
define Amax                 = 48
define Anow                 = 1
define Astep                = 0.025
define Nx                   = 1024
define Ny                   = 1024
define Nz                   = 1024
define Nspace               = 1
define Theta                = 0.3
define Nstep                = 1881
define Stepcount            = 1
define Stepnum              = 1
define Iseed                = -56
define Powerflag            = 2
define Ascii Powerfile      = camb.z=47_0.31_-1.2_-2.4.ascii
define rvfilename           = INITIAL
define rvprefix             = INITIAL
#End of Ascii Header
```

14.2 Compilation

Rules.make Settings

```
CC = mpiicx
FC = mpiifx
OPT = -DINTEL -O2 -qopenmp
CDFLAGS = -DWGROUPSIZE=80 -DNMEG=8000L -DINCLUDE_TREE_FORCE \
          -D_LARGE_FILES -DSAVESLICE -DPMSEEDFORCE
COMFLAGS = -DINDEX -DVarPM -DXYZDBL -DMPI_SLOW
FFTWDIR = /home/kjhan/local/
```

14.3 Execution

With 16 MPI ranks and 4 OpenMP threads per rank (64 cores total):

```
export OMP_NUM_THREADS=4
srun --ntasks=16 --cpus-per-task=4 ./namu.exe params.dat
```

14.4 Expected Output

The simulation starts at $z = 47$ ($a = 1$) and evolves to $z = 0$ ($a = 48$) in 1881 steps. Key output messages:

- `Poisson Correction factor X at anow = Y` — Dark energy correction at each step.
- `CPU(F-FFTW3)` and `CPU(B-FFTW3)` — Forward/backward FFT timing.
- `Treecorrectingtime CPU= X` — Tree force correction timing.
- `Step N CPU= X` — Total time per step.